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ORIGINAL PAPER

Organically modified sodium montmorillonite for enhanced adsorption of xylenol orange from aqueous solutions: kinetic, thermodynamic, and mechanistic study

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Abstract Xylenol orange (XO) is a persistent anionic azo dye that resists conventional treatment. Cetyltrimethylammonium bromide (CTAB) and binding poly (vinylpyrrolidone) (PVP) were sequentially intercalated onto Na-montmorillonite (NaMt) to prepare a CTAB/PVP-modified Na-montmorillonite (CP-NaMt), and its adsorption toward XO was evaluated. The nanostructured adsorbent was characterized by chemical analysis, morphological, phase analysis, and point-of-zero-charge measurements. The characterization indicated successful organic intercalation accompanied by slight basal-spacing expansion, partial layer exfoliation and the development of a hierarchical micro-mesoporous structure. The Brunauer–Emmett–Teller specific surface area increased from 46.5 to 88.7 $m^2 \cdot g^{-1}$. Batch tests identified pH = 5 as optimal, consistent with XO speciation and the positively charged surface of CP-NaMt. The adsorption process reached equilibrium rapidly, with a maximum capacity of 93.85 $mg \cdot g^{-1}$ achieved within 40 min at 30 °C. Kinetics followed a pseudo-second-order model and intraparticle-diffusion analysis indicated multistage transport beginning with a film-diffusion regime. Equilibrium isotherms were better described by the Langmuir model. Thermodynamic analysis indicated a spontaneous, endothermic process with an entropy gain that facilitates desolvation and mass transport. Regeneration with ethanol maintained ≥ 87.4% removal over five cycles. Spectroscopic evidence and control experiments supported a mechanism dominated by electrostatic ion pairing at quaternary ammonium sites, further stabilized by hydrogen bonding and hydrophobic partitioning and π-π stacking within organic microdomains. This low-cost, regenerable clay platform shows potential for polishing dye-laden wastewater.

Keywords Adsorbent · Anionic azo dye · Cetyltrimethylammonium bromide · Equilibrium isotherms · Wastewater

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Introduction

Water scarcity and pollution have become global environmental challenges. Recent research indicates that over 80% of industrial and domestic wastewater is discharged without proper treatment, leading to significant ecological damage and posing direct threats to human health (Magro et al., 2020). Water pollutants are mainly categorized into organic contaminants (such as industrial dyes, pesticide residues, and antibiotics) and inorganic pollutants (such as heavy metal ions including Cd, Cu, Ni, Pb, and As) (Ali et al., 2023; Jioui et al., 2023; Loukili et al., 2021). Among these, xylenol orange (XO), a representative azo dye, is widely used in textile dyeing and chemical analysis due to its excellent metal ion chelating properties (Gecol, 2004; Meriç et al., 2005). However, this compound exhibits significant environmental risks, including high toxicity, resistance to degradation, and potential carcinogenicity (Al-Kazragi et al., 2019; Benkhatou et al., 2025). Studies have shown that XO can reduce the photosynthesis efficiency of aquatic plants by 40–60% and bioaccumulate in higher organisms through the food chain, ultimately harming human health (Alzain et al., 2023). Moreover, due to the presence of sulfonic groups in its molecular structure, XO demonstrates strong solubility and chemical stability in water, and traditional water treatment methods often achieve less than 30% removal efficiency (Pal et al., 2025).

Thus, developing efficient, cost-effective, and environmentally friendly XO removal technologies is crucial for improving water quality and protecting aquatic life. Existing XO removal methods include chemical oxidation (Abuzaytoun et al., 2020), photocatalytic degradation (Xu et al., 2024), and adsorption (Benkhatou et al., 2025). Although methods like chemical oxidation and photocatalysis show promising results under specific conditions, their high energy consumption, elevated costs, and potential toxic by-products limit their large-scale application. In contrast, adsorption methods, known for their simplicity, low cost, and environmental friendliness, offer significant value in organic dye wastewater treatment (Samanta et al., 2024; Yu et al., 2018). Chen et al. (2012) indicated that MIL-101(Cr), a metal–organic framework (MOF), achieves up to 97.2% XO removal from an initial concentration of 100 $mg \cdot L^{-1}$.

In recent years, montmorillonite (Mt), a natural clay mineral, has attracted widespread attention as an adsorbent in water treatment due to its abundant availability, low cost, excellent adsorption properties, and environmental friendliness (Joudi et al., 2022; Sarkar et al., 2024). However, natural Mt is often limited by its inherent adsorption capacity and selectivity, requiring modification to enhance its pollutant removal performance. Sodium ion exchange modification effectively converts raw Mt into Na-Mt, significantly enhancing its adsorption performance (Peng et al., 2025). To further optimize the adsorption performance of Na-Mt, researchers have developed various surface modification methods.

Cetyltrimethylammonium bromide (CTAB), a cationic surfactant, enters the interlayers of Mt through ion exchange (Boutaleb et al., 2021). The long alkyl chains of CTAB not only expand the interlayer spacing but also make the clay surface hydrophobic, enhancing its affinity for organic pollutants (Moja et al., 2020). Additionally, polymeric materials such as polyvinylpyrrolidone (PVP), polyvinyl alcohol (PVA), and polyaniline (PAni) contain characteristic functional groups (carbonyl, hydroxyl, and imino groups) within their molecular structures (Krasinskyi et al., 2023). These groups can form hydrogen bonds, van der Waals forces, and coordination interactions with pollutants, significantly improving the materials' adsorption performance (Bhuyan, M. M. et al., 2025). Mennas et al. (2023) confirmed that CTAB modification alone increased the adsorption capacity of Mt for methylene blue from 57 mg g^{-1} (in its natural state) to 81 mg g^{-1}, and further modification by combining CTAB with conductive polymer PAni enhanced the adsorption capacity to 109 $mg \cdot g^{-1}$. In this context, this study adopts a synergistic modification approach, incorporating CTAB and PVP into the Na-Mt structure to develop an advanced material, CP-NaMt, with improved performance. The synergistic modification using CTAB and PVP aims to address the limitations of single modifications, fully leveraging the advantages of both.

The main objectives of this study are as follows: (1) to establish a simple method for fabricating CP-NaMt and perform characterization using techniques such as Fourier transform infrared spectroscopy (FT-IR), X-ray diffraction (XRD), scanning electron

microscopy (SEM), BET surface area analysis, and zeta potential measurement; (2) to investigate the optimal adsorption conditions, kinetics, isothermal behavior, and thermodynamic properties of CP-NaMt for XO removal; and (3) to explore the interaction mechanisms between CP-NaMt and XO.

Materials and methods

Materials and reagents

Anhydrous sodium carbonate (Na_2CO_3, ≥ 99.8%), Sodium hydroxide (NaOH, ≥ 99%), hydrochloric acid (HCl, 37%) and anhydrous ethanol (C_2H_5OH, ≥ 99.7%) were purchased from Sinopharm Chemical Reagent Co., Ltd. (Shanghai, China). Cetyltrimethylammonium bromide ($C_{16}H_{33}N(CH_3)_3Br$, ≥ 99%, Macklin, China) and polyvinylpyrrolidone (K30, Mw≈40,000, Sigma-Aldrich, USA) were used. Ultrapure water (resistivity ≥ 18.2 MΩ·cm) was produced using a Milli-Q water purification system. Xylenol Orange was purchased from Tianxin Chemicals Co., Ltd. (Tianjin, China). Additionally, the raw montmorillonite was purchased from Aladdin Biochemical Technology Co., Ltd. (Shanghai, China).

Preparation of CP-NaMt

The synthesis of modified montmorillonite was carried out in three main steps (Fig. 1):

Purification of Mt: The raw Mt was ground into fine powder, sieved through a 200-mesh screen, and dispersed in ultrapure water. After settling for 1 min, the supernatant was decanted. This procedure was repeated three times, and the combined suspensions were allowed to settle for 5 h. The precipitate was then collected and dried at 80 °C to obtain purified Mt.

Preparation of Na-Mt: A mixture of 2.5 g of purified Mt and 0.25 g of anhydrous Na_2CO_3 was dispersed in 50 mL of ultrapure water. The suspension was subjected to ultrasonic treatment for 15 min, followed by stirring at 60 °C for 5 h. The product was washed five times with ultrapure water and once with anhydrous ethanol, then vacuum-dried at 60 °C for 12 h to obtain Na-Mt.

Dual Modification with CTAB and PVP: The sample of Na-Mt was dispersed in deionized water (solid-to-liquid ratio 1:40) and stirred for 30 min. Then 1 g of CTAB was added and the mixture was stirred at 40 °C for 3 h to facilitate interlayer adsorption. Afterward, 1 g of PVP was introduced and the reaction continued at 40 °C for an additional 3 h. The product was centrifuged at 11,000 rpm for 3 min, thoroughly washed three times with deionized water, dried at 80 °C for 24 h. Finally, the modified adsorbent was passed through a 100-mesh sieve to obtain the uniform CP-NaMt composite.

Characterization of the modified adsorbent

The microstructure and surface morphology were analyzed by scanning electron microscopy (SEM, Hitachi SU8010, Japan) at an accelerating voltage of 5–15 kV. The surface functional groups were identified by Fourier-transform infrared spectroscopy (FT-IR, Thermo Fisher Scientific Nicolet 6700, USA) using the KBr pellet transmission method (400–4000

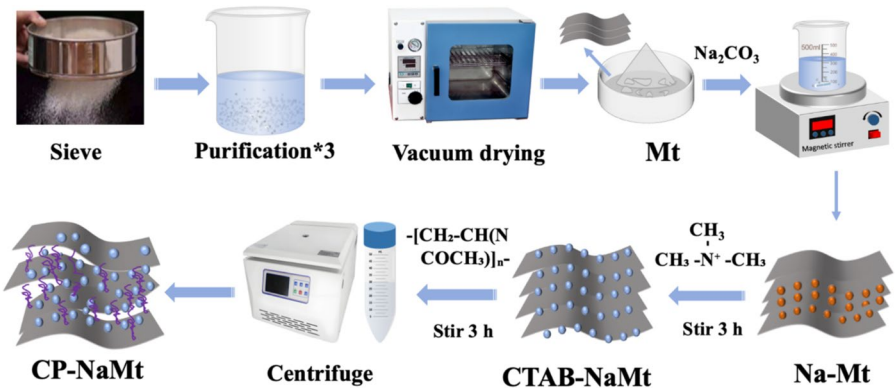

Fig. 1 Flowchart for the preparation of CP-NaMt

cm^{-1}, 4 cm^{-1} resolution, 32 scans). N_2 adsorption–desorption isotherms were measured at 77 K (BET, Micromeritics ASAP 2460, USA) to determine specific surface area, pore volume and pore size distribution after degassing samples at 120 °C for 6 h. The zeta potential as a function of pH was measured using a zeta potential and particle size analyzer (90Plus PALS, USA). The crystal structure was analyzed by X-ray diffraction (XRD, Rigaku RINT2000, Japan) using Cu-Kα radiation (λ = 1.5404 Å), with measurements conducted over a 2θ range of 5–80° at a scanning rate of 4° min^{-1}. The interlayer spacing value (d-spacing) was determined from the Bragg's formula (Eq. (1)) (Boutaleb et al., 2021):

$$d = \frac{\lambda}{2\sin\theta} \tag{1}$$

where λ is the X-rays wavelength and θ is the diffraction angle.

Adsorption experiments

The adsorption capacity of XO in aqueous solution was evaluated through triplicate batch experiments. Specifically, 50 mL XO solutions (10–200 mg L^{-1}, pH 1–9 adjusted with 0.1 M HCl/NaOH) were incubated with a predetermined adsorbent dose at controlled temperatures (10–50 °C) on a mechanical shaker (150 rpm). At designated time intervals (5–60 min), samples were filtered through 0.45 μm filter membranes. The residual XO concentration was measured at 580 nm using a UV–Vis spectrophotometer (T80, PG Instruments, UK), with calibration standards and blank corrections applied. The removal efficiency (R, %) and the adsorption capacity (q_e, mg g^{-1}) of XO were calculated by the following Eq. (2) and Eq. (3) (Salah Omer et al., 2022):

$$R = \frac{C_0 - C_e}{C_0} \times 100\% \tag{2}$$

$$q_e = \frac{V(C_0 - C_e)}{m} \tag{3}$$

where C_o is the initial XO concentration and C_e is the equilibrium XO concentration in mg L^{-1}; while q_e is the amount of anionic dye adsorbed after equilibrium in mg g^{-1}. V is the sample volume (L) and W is the weight of the adsorbent (g).

Adsorption kinetic, isotherm and thermodynamic models

The adsorption kinetics were analyzed using the non-linear pseudo-first-order (PFO) and pseudo-second-order (PSO) models. The PFO model (Eq. (4)) is written as:

$$q_t = q_e\left(1 - e^{-k_1 t}\right) \tag{4}$$

and the PSO model (Eq. (5)) as:

$$q_t = \frac{k_2 q_e^2 t}{1 + k_2 q_e t} \tag{5}$$

To further elucidate the mass-transfer mechanism, the intra-particle diffusion model of Weber-Morris (Eq. (6)) was applied (Rathore et al., 2016):

$$q_t = k_d t^{0.5} + C \tag{6}$$

where q_e and q_t are the amounts of XO adsorbed at equilibrium and at contact time (t), k_1 and k_2 are the pseudo-first order rate constant (min^{-1}) and pseudo-second order rate constant (g mg^{-1} min^{-1}), respectively. K_d is the intra-particle diffusion rate constant (mg g^{-1} $min^{0.5}$) and C (mg g^{-1}) is the intercept associated with the boundary-layer effect.

The equilibrium adsorption of XO onto CP-NaMt was analyzed using the non-linear forms of the Langmuir and Freundlich isotherm models. The Langmuir model (Eq. (7)) assumes monolayer coverage on a homogeneous surface and is expressed as:

$$q_e = \frac{q_{max} K_L C_e}{1 + K_L C_e} \tag{7}$$

The favorability of adsorption was further evaluated using the dimensionless Langmuir separation factor R_L, defined as (Eq. (8)):

$$R_L = \frac{1}{1 + K_L C_0} \tag{8}$$

The Freundlich model (Eq. (9)) describes adsorption on a heterogeneous surface and is given by (Foo & Hameed, 2010):

$$q_e = K_F C_e^{1/n} \tag{9}$$

where K_L is the Langmuir isotherm coefficient (L mg^{-1}), q_{max} is the Langmuir constant (mg g^{-1}) connected to the maximum monolayer adsorption capacity, K_F is the Freundlich isotherm coefficient (L g^{-1}),

1/n is a function of the strength of the adsorption, n is the adsorption intensity. When n = 1, adsorption is linear; when n < 1, adsorption is unfavorable. A value of 1 < n < 10 indicates favorable adsorption (Zhang et al., 2019). For the Langmuir separation factor R_L, When $R_L > 1$ indicates unfavorable adsorption, R_L = linear adsorption, $0 < R_L < 1$ favorable adsorption, $R_L = 0$ irreversible adsorption (Sharma & Naushad, 2020).

The thermodynamic parameters were calculated based on the Langmuir constant K_L. For isotherms expressed in mg g^{-1} and mg·L^{-1}, the dimensionless equilibrium constant K_c was obtained by converting the Langmuir constant K_L from mg^{-1} to L mol^{-1} through multiplication by 10^3 M (Tran, 2022). Thermodynamic parameters are Gibbs energy (ΔG, kJ mol^{-1}), enthalpy change (ΔH, kJ mol^{-1}), and entropy change (ΔS, J mol^{-1} K^{-1}), which were calculated by Gibbs–Helmholtz equation (Eq. (10)) and Van't Hoff equation (Eq. (11)) (Ahmad et al., 2016).

$$\Delta G^\circ = -RT \ln K_c \tag{10}$$

$$\ln K_c = -\frac{\Delta H^\circ}{RT} + \frac{\Delta S^\circ}{R} \tag{11}$$

where M is the molecular weight of XO (g mol^{-1}), R is the gas constant (8.314 J mol^{-1} K^{-1}), Kc is the partition coefficient (L·mol^{-1}).

Preparation of standards and calibration curve

A 1000 mg·L^{-1} stock solution of XO was prepared in deionized water and stored at 4 °C in the dark. Working standards (0–200 mg L^{-1}) were freshly prepared by serial dilution immediately before measurement. A reagent blank (deionized water) was used for baseline correction. Absorbance was recorded at the XO absorption maximum under the same matrix and pH as the samples using 1.0 cm quartz cuvettes on a UV–Vis spectrophotometer (U-2900, Hitachi, Japan). The calibration curve for XO showed excellent linearity over 0–200 mg L^{-1} ($y = 0.0200x + 0.0194$, $R^2 = 0.9996$), confirming that a single linear regression model is suitable for quantitative analysis (see Fig. S2).

Results and discussion

Adsorbent characterization and evaluation

The morphological and structural evolution of the materials before and after modification was analyzed by SEM at magnifications of 2,000× and 10,000×. As shown in Fig. 2a, the raw montmorillonite exhibits a typical layered spherical morphology, composed of tightly stacked plate-like crystal layers (Wang et al., 2024). The particle surface is relatively smooth and dense. In contrast, the modified CP-NaMt forms looser aggregates with partially exposed lamellar edges and a rougher, more porous surface, as shown in Fig. 2b (Mennas et al., 2023). These SEM observations suggest a loosening of the stacked layers and partial exfoliation of the Mt particles after CTAB-PVP modification, consistent with the increase in BET surface area and is expected to favor XO adsorption.

Figure 3 further compares XRD patterns of the material before and after CTAB/PVP modification. Both materials show characteristic diffraction peaks of montmorillonite, including reflections from the (001), (020)/(110), (021), (220), and (060) crystal planes, with a quartz impurity peak at (101), confirming that the sample is purified natural montmorillonite (Table S1) (Huff, 1990). After CTAB/PVP modification, the low-angle (001) basal reflection becomes broader and less intense, appearing as a plateau rather than a sharp peak. The center of this plateau shifts from 5.92° (d001 = 1.49 nm) for Na-Mt to about 5.79° (d001 = 1.53 nm) for CP-NaMt, indicating a slightly larger average basal spacing and a range of interlayer distances caused by partial intercalation and local exfoliation. Notably, although long-chain surfactant intercalation typically leads to significant expansion of clay interlayer spacing (Δd = 0.3 nm) (Guo et al., 2025; Mennas et al., 2023), the observed expansion in this study (Δd = 0.04 nm) is much smaller. This modest change likely reflects partial intercalation and structural rearrangement under CTAB/PVP treatment, while preserving the layered framework (Wang et al., 2024). CTAB cations partially displace interlayer water to form tightly packed monolayers. Simultaneously, PVP stabilizes the organic ions through hydrogen bonding while promoting water expulsion, ultimately generating dense hydrophobic

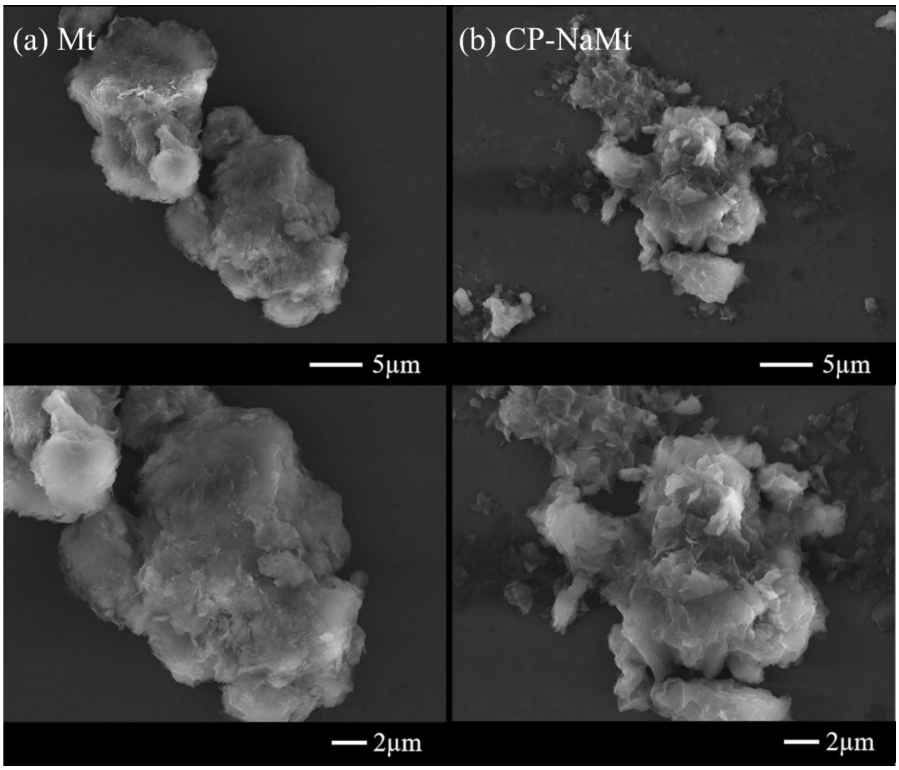

Fig. 2 The SEM images of Mt **a** and CP-NaMt **b** at magnifications of 2,000× and 10,000×

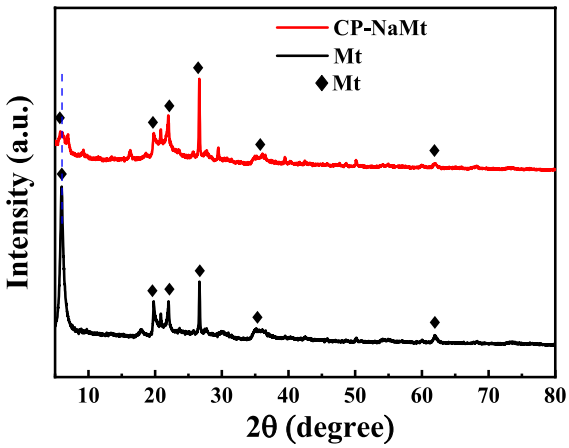

Fig. 3 The XRD patterns of Mt and CP-NaMt

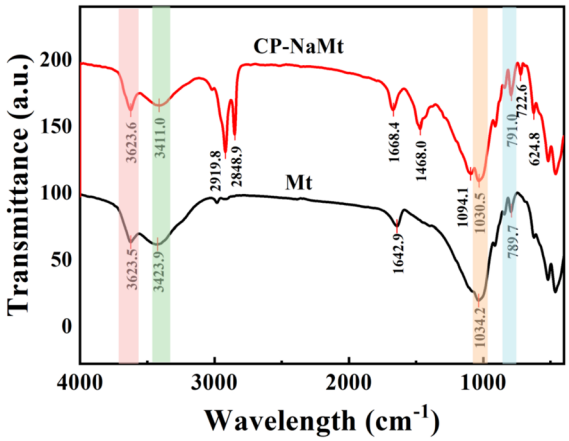

Fig. 4 The FT-IR spectra of Mt and CP-NaMt

microdomains (Zhu et al., 2019). The synergistic effect preserves the crystalline framework while promoting nanosheet exfoliation. As confirmed by SEM, this structural evolution generates a porous wrinkled architecture that provides abundant external surfaces and mesoporous sites for efficient adsorption of macromolecular pollutants.

FT-IR analysis (Fig. 4) reveals the chemical structural evolution of montmorillonite before and after modification. The raw sample shows typical Al–OH internal stretching vibrations at 3623 cm^{-1} and interlayer water O–H stretching vibrations at 3439 cm^{-1}, with broad and intense peaks confirming the material's surface is rich in hydroxyl groups and highly

hydrated (Bensedira et al., 2022). After Na^+ pretreatment and CTAB/PVP modification, the interlayer water vibration peak shifts significantly from 3439 cm^{-1} to 3411 cm^{-1}. This shift confirms that hydrophobic CTAB chains shield surface hydroxyl groups via steric hindrance while simultaneously driving the expulsion of interlayer water through organic intercalation (Wang et al., 2024; Zhang et al., 2014). New absorption peaks at 2919 cm^{-1} and 2848 cm^{-1} correspond to the asymmetric and symmetric stretching vibrations of the -CH_2 group in CTAB, directly confirming the successful anchoring of CTAB within the montmorillonite layers and surface (Mennas et al., 2023). Furthermore, the simultaneous appearance of characteristic peaks at 1668 cm^{-1} (C=O stretching), 1468 cm^{-1} (CH_2 deformation), and 1094 cm^{-1} (C–O vibration) provides fingerprint evidence of PVP molecules binding to montmorillonite (Krasinskyi et al., 2023). Notably, the Si–O–Si skeletal vibration peak shifts from 1034 cm^{-1} to 1030 cm^{-1} with a narrowing of the peak, reflecting the expansion of interlayer spacing induced by organic intercalation (Boutaleb et al., 2021). This result is consistent with the interlayer expansion observed in the XRD measurements, providing direct evidence of the modification effect.

The textural properties of the composites were characterized by N_2 adsorption–desorption isotherms and BJH pore-size distribution analysis before and after modification. As shown in Fig. S1a and Fig. S1b, the pristine Mt exhibits a type IV isotherm with an H_3 hysteresis loop, consistent with the International Union of Pure and Applied Chemistry (IUPAC) classification system for porous materials (Thommes et al., 2015). The limited low-pressure adsorption ($P/P_0 < 0.1$) and 46.5 $m^2\ g^{-1}$ BET surface area, along with a narrow BJH distribution peaking at approximately 2–3 nm, reflect microporous dominance (Daas & Zaghouane-Boudiaf, 2022). After dual modification, CP-NaMt (Fig. S1c, Fig. S1d) shows pronounced textural optimization, with adsorption increasing across the entire pressure range and a broader hysteresis loop emerging at high relative pressure ($P/P_0 > 0.8$). The BET surface area rises to 88.7 $m^2 \cdot g^{-1}$, representing a 91% increase. The pore-size distribution becomes bimodal, retaining micropores while introducing additional 10–30 nm mesopores, accompanied by a marked increase in total pore volume. This hierarchical porosity originates from the expansion of interlayer spacing, exfoliation of lamellae, and subsequent reassembly into a meso-macroporous network during the organic–inorganic synergistic modification process (Wang et al., 2024). These results are consistent with the wrinkled, porous lamellar morphology observed by SEM and with the interlayer expansion evidenced by XRD.

Effect of pH on XO adsorption

The pH dependence of XO adsorption on CP-NaMt was systematically evaluated (Fig. 5c). Adsorption experiments were carried out at 30 °C by contacting 0.05 g of adsorbent with 50 mL of XO solution at an initial concentration of 100 mg L^{-1}. As the pH increases from 2 to 5, the equilibrium capacity rises sharply, reaches a maximum around pH = 5, and then declines slightly under alkaline conditions (pH > 5).

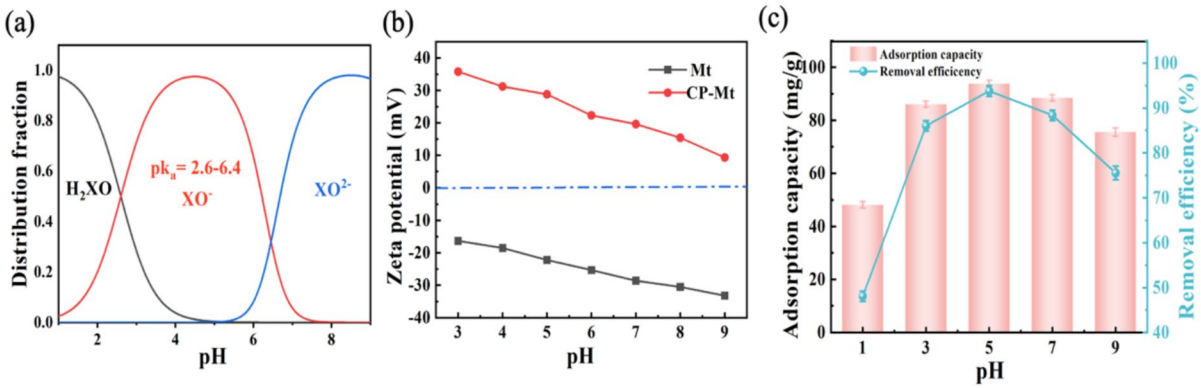

Fig. 5 The Zeta potential of Mt and CP-NaMt **a**; The fraction of XO species at different pH **b**; Effect of pH value on XO by CP-NaMt **c**

This trend is governed by XO speciation (Fig. 5a) and the surface charge of the adsorbent (Fig. 5b). The speciation analysis, based on reported pK_a values (2.6 and 6.4), confirms XO^- as the predominant chemical species at pH = 5 (Liosi et al., 2017). Electrostatic attraction between XO^- and the quaternary-ammonium sites on CTAB ($-N^+(CH_3)_3$) promotes binding. In addition, hydrophobic interactions between CTAB alkyl chains and the aromatic rings of XO, together with hydrogen bonding between PVP carbonyl groups and phenolic -OH on XO, act synergistically to maximize adsorption near pH = 5 (Irfan et al., 2014; Mohammad et al., 2021). At lower pH (< 5), protonation of acidic groups on XO diminishes its anionic character and weakens electrostatic attraction. At higher pH (> 5), competition from OH^-, progressive conversion to the more highly charged XO^{2-}, and the decrease in the positive zeta potential of CP-NaMt collectively reduce adsorption. Therefore, pH = 5 was identified as the optimal condition for XO adsorption.

Effect of adsorbent dosage on XO adsorption

The interlayer galleries and polar functional groups of CP-NaMt favor XO uptake, so adsorbent dosage is a critical parameter. As shown in Fig. S3, at pH = 5 and an initial concentration of 100 $mg \cdot L^{-1}$, 0.05 g of adsorbent in 50 mL of XO solution yielded an equilibrium capacity of approximately 94 $mg \cdot g^{-1}$. Increasing the dosage further improved the overall removal efficiency but decreased the capacity per unit mass because the residual XO concentration fell and a larger fraction of active sites remained unused, a phenomenon often termed capacity dilution (Abdallah & Alprol, 2024). At still higher solid–liquid ratios, particle agglomeration can occur, reducing the accessible surface area and suppressing additional gains in adsorption. Considering both efficiency and capacity, 0.05 g represents the optimal dosage for this system.

Contact time and kinetic modeling

Contact time is critical for diagnosing the rate at which hazardous pollutants are captured and removed. All kinetic experiments were conducted by contacting 0.05 g of CP-NaMt with 50 mL of XO solution at an initial concentration of 100 $mg \cdot L^{-1}$ and pH = 5, with initial concentrations 100 mg L^{-1}. The influence of contact time on XO adsorption by CP-NaMt was examined from 3 to 60 min (Fig. 6a). A rapid uptake occurred within the first 10 min, reaching > 80% of the equilibrium capacity, and equilibrium was essentially achieved by approximately 40 min. The initial acceleration reflects the high abundance and accessibility of surface-active sites, whereas the rate decreases as adsorption proceeds due to declining site availability and increasing surface repulsion (Açıkyıldız et al., 2023).

To elucidate the rate-controlling step, the kinetic data were fitted with the PFO model and the PSO model. The kinetic data of XO are shown in Fig. 5b, Fig. 5c and Table 1. The PFO and PSO models are empirical rate expressions; a better fit to the PSO form does not by itself prove chemisorption, but often reflects site limited uptake kinetics (Zhou et al., 2021). Both kinetic models described the data well ($R^2 > 0.96$). However, the PSO model gave slightly higher coefficients of determination

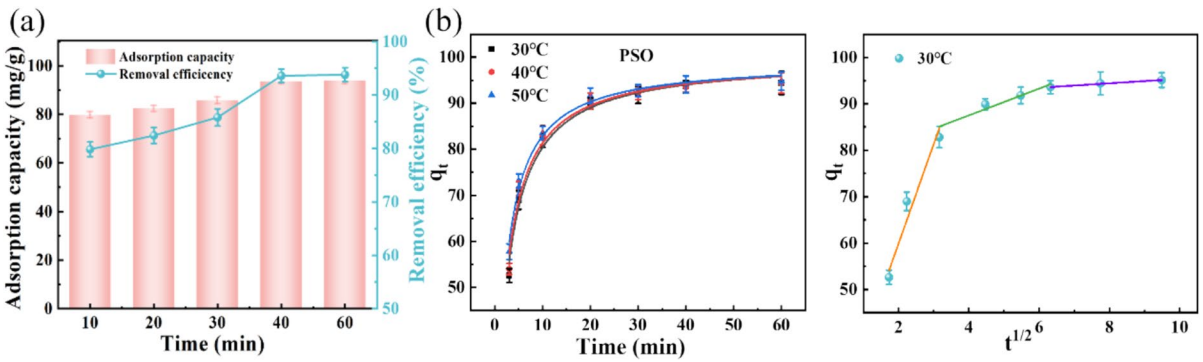

Fig. 6 Effect of time on XO by CP-NaMt **a**; PSO kinetic model **b**; Weber-Morris model plots for XO adsorption by CP-NaMt **c**

Table 1 Kinetic parameters of CP-NaMt adsorption XO

T	PFO model			PSO model			Real value
	q_e (mg g^{-1})	K_1 (min^{-1})	R^2	q_e (mg g^{-1})	K_2 ($g{\cdot}mg^{-1}{\cdot}min^{-1}$)	R^2	
30 ℃	92.05 ± 1.0	0.27 ± 0.01	0.9844	99.65 ± 1.3	0.0042 ± 0.0003	0.9867	93.80
40 ℃	92.19 ± 1.1	0.28 ± 0.01	0.9805	99.38 ± 1.4	0.0046 ± 0.0004	0.9830	94.45
50 ℃	92.53 ± 1.2	0.31 ± 0.02	0.9640	99.66 ± 1.0	0.0051 ± 0.0003	0.9796	94.87

(R^2 = 0.980–0.987) and a calculated equilibrium capacity 99 mg·g^{-1} that closely matches the experimental value 94 mg·g^{-1}. The pseudo-second-order curves appear close to each other because equilibrium is reached rapidly in all cases and the equilibrium capacities only vary within a narrow range. The main temperature effect is reflected in the rate constant K_2 rather than in q_e. These results indicate that chemisorption-like uptake governed by electrostatic ion-pairing at quaternary-ammonium sites is rate-controlling.

Building on the kinetic fits that favored the PSO model and suggested chemisorption-controlled adsorption, we further examined the mass-transfer steps. The kinetic data at 30 °C were analyzed using the Weber-Morris intraparticle-diffusion model to distinguish the contributions of external film diffusion and intraparticle diffusion. The Weber-orris intraparticle diffusion model results are shown in Fig. 6c and Table S2. This analysis revealed a three-stage transport sequence with apparent rate constants K_1 = 20.34, K_2 = 3.35, K_3 = 0.47, corresponding to film diffusion, pore diffusion, and surface coordination toward equilibrium, respectively. The multilinear plots do not pass through the origin, confirming the coexistence of boundary-layer resistance with intraparticle diffusion. The enlarged surface area and hierarchical porosity of CP-NaMt shorten both film and intraparticle-diffusion paths, thereby imparting rapid and controllable adsorption kinetics.

Initial concentration effect and isotherm modeling

The initial dye concentration is a key determinant of adsorption performance. As the initial XO concentration increased from 50 to 200 mg·L^{-1}, the equilibrium capacity rose progressively (Fig. 7a). These isotherm experiments were carried out at 30 °C and pH 5 by contacting 0.05 g of CP-NaMt with 50 mL of solution for 40 min. A higher C_0 provides a larger concentration gradient and drives more XO molecules to occupy available adsorption sites. However, once a substantial fraction of sites is occupied, the incremental gain in q_e diminishes as the surface approaches saturation.

Adsorption isotherms for XO on CP-NaMt were fitted with the Langmuir and Freundlich model. The Langmuir isotherm assumes monolayer adsorption on a homogeneous surface with energetically equivalent sites, whereas the Freundlich isotherm is an empirical equation that represents adsorption on heterogeneous surfaces without an explicit monolayer capacity (Anastopoulos et al., 2016). The adsorption isotherm parameters and corresponding

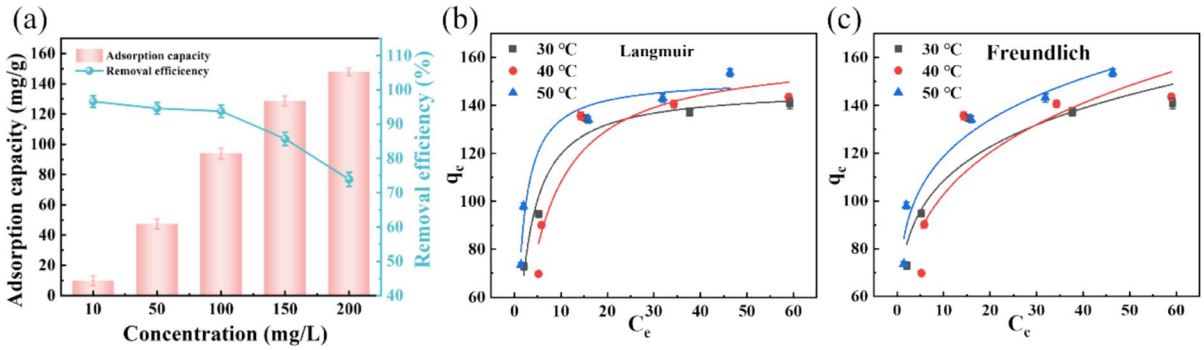

Fig. 7 Effect of concentration on XO by CP-NaMt **a**; Langmuir model **b**; Freundlich model 0 for XO adsorption by CP-NaMt **c**

R^2 coefficients for both Langmuir and Freundlich models are presented in Fig. 7b, Fig. 7c and Table 2. The experimental data showed better agreement with the Langmuir model, as evidenced by higher correlation coefficients ($R^2 = 0.997–0.999$). While the Freundlich model yielded slightly lower fitting quality, its parameters still suggested favorable adsorption conditions, with the heterogeneity factor $n > 1$ and the Freundlich constant K_F increasing with rising temperature (Belhachemi & Addoun, 2011; You et al., 2015). Further analysis using the Langmuir separation factor R_L revealed exceptionally low values (3.6×10^{-2}–3.1×10^{-2}) across the studied temperature range (30–50 °C), indicating the highly favorable nature of the adsorption process (Sharma & Naushad, 2020; Sherlala et al., 2019). Although CP-NaMt contains heterogeneous domains that could support multilayer uptake, the isotherms over 30–50 °C were better described by the Langmuir model (Al-Kazragi et al., 2019). The evidence collectively indicates that XO adsorption occurs predominantly as a monolayer at CTAB-derived quaternary-ammonium sites on CP-NaMt, primarily governed by electrostatic ion-pairing interactions.

Adsorption temperature and thermodynamics

To clarify the influence of operating conditions, the effect of temperature was examined (Fig. S4). The highest removal efficiency of XO occurred at 30 °C, and the efficiency remained essentially constant across 30–50 °C (93.85–95.42%) under otherwise identical conditions ($C_0 = 100$ mg L^{-1}, pH 5, 0.05 g CP-NaMt in 50 mL of solution, 40 min contact time). Higher temperatures lower the diffusion barrier for XO through interlayer galleries and strengthen interfacial interactions between XO and the active sites on the modified montmorillonite, which together enhance the adsorption capacity (Fouda & Hassan, 2024).

The thermodynamic results were shown in Fig. S6 and Table S3. The ΔG^0 was determined to range from − 28.54 to − 33.98 kJ·mol^{-1} across the temperature range of 30–50 °C. These negative ΔG° values confirm the spontaneous nature of the adsorption process, with increasing spontaneity at elevated temperatures (Lyu et al., 2022). Further analysis through Van't Hoff plots revealed an endothermic enthalpy change ΔH^0 of + 34.29 kJ·mol^{-1} and a positive entropy change ΔS^0 of + 212.23 J·mol^{-1}·K^{-1}. These thermodynamic parameters demonstrate that heating facilitates the desolvation process, increases molecular mobility, and strengthens interfacial interactions, collectively leading to enhanced adsorption capacity (Benkhatou et al., 2025).

Comparison of adsorption effect of XO by different adsorbents

To assess adsorption performance, we benchmarked CP-NaMt against reported adsorbents for XO removal under comparable conditions. Table 3 summarizes their performance across three key types of materials. Conventional mineral wastes (e.g., coal ash) require high dosages (20 g·L^{-1}) yet show low capacity (0.74 mg·g^{-1}) and only 80% removal at $C_0 = 30$ mg·L^{-1}, indicating their weak affinity for XO. Functionalized materials (MIP, H-Mag/Aw, BXT-HDTMA) operate at lower dosages (0.1–0.8 g·L^{-1}) and can yield high efficiencies (up to 98%), but their mass-specific capacities are modest (3.43–24.4 mg·g^{-1}) and susceptible to solution conditions. Polymeric/biopolymer systems (PEI-grafted chitosan microspheres) exhibit very high capacity (306 mg·g^{-1}) and 95.6% removal, albeit at a much higher influent concentration ($C_0 = 400$ mg·L^{-1}). By contrast, CP-NaMt achieves 93.85% removal with a capacity of 93.85 mg·g^{-1} at pH = 5, $C_0 = 100$ mg·L^{-1}, and a dosage of 1 g·L^{-1}, providing a favorable balance of low dosage, high efficiency, and competitive capacity under moderate conditions. These findings indicate CP-NaMt

Table 2 Isotherm parameters of XO adsorption by CP-Mt

T	Langmuir				Freundlich		
	q_{max} (mg g^{-1})	K_L (L mg^{-1})	R^2	R_L	K_F (L g^{-1})	n	R^2
30 ℃	147.37 ± 4.9	0.43 ± 0.1	0.9678	0.036	71.89 ± 9.4	5.61 ± 1.3	0.8750
40 ℃	159.54 ± 10.8	0.31 ± 0.1	0.9008	0.034	63.76 ± 11.3	4.59 ± 1.5	0.7655
50 ℃	151.40 ± .5.1	0.76 ± 0.1	0.9638	0.031	79.13 ± 6.4	5.69 ± 0.8	0.9483

Table 3 Comparison of CP-NaMt for XO adsorption with other adsorbents in the references

Adsorbents	pH	Initial concentration (mg L^{-1})	Adsorbent dosages (g L^{-1})	Removal efficiency (%)	Adsorption capacity (mg g^{-1})	References
Coal ash	2	30	20	80	0.74	(Ishaq et al., 2014)
MIP	7	10	0.7	80	12	(Bhawani et al., 2020)
H-Mag/Aw	7.11	50	0.8	61	24.4	(Benkhatou et al., 2025)
BXT-HDTMA	9.03	10	0.1	98	3.43	(Al-Kazragi et al., 2019)
PEI-grafted chitosan microspheres	4	400	1.25	95.6	306	(Yu et al., 2018)
CP-NaMt	5	100	1	93.85	93.85	This work

promising potential for practical application in treating water contaminated with low-to-moderate XO concentrations. It is worth noting that the tested XO concentrations (up to 200 mg·L^{-1}) and near-ambient operating temperatures are consistent with those encountered in textile and analytical dye effluents, indicating that the evaluated performance of CP-NaMt can be directly extrapolated to realistic wastewater treatment scenarios (Al-Kazragi et al., 2019; Chen et al., 2012).

Recyclability of adsorbents

To evaluate practical reusability, we conducted five consecutive adsorption–desorption cycles at a fixed solid–liquid ratio of 1 g·L^{-1}, employing two distinct regeneration protocols: 95% ethanol treatment (25 °C, 30 min) and a mixed solution of 0.05 M NaOH and 0.1 M NaCl (40 °C, 30 min) (Akl et al., 2023; Alguacil & López, 2021; Momina et al., 2019). The results are shown in Fig. S6. Remarkably, CP-NaMt retained 87.4% and 84.5% removal efficiency after five regeneration cycles with ethanol and alkaline-salt treatments, respectively. The minimal 2.9% performance difference suggests comparable regeneration efficacy between these methods. These findings collectively confirm the excellent reusability and regeneration potential of CP-NaMt for practical applications. From an environmental and practical viewpoint, this favorable reusability means that a single batch of CP-NaMt can be used repeatedly, thereby helping to limit both chemical consumption and the amount of secondary solid waste generated. After use, the exhausted adsorbent is essentially a low-volume clay-rich solid loaded with XO, which can be collected and handled as a process residue within existing solid-waste management systems rather than being discharged directly to receiving waters.

Adsorption mechanism

The proposed synergistic adsorption mechanism of XO on CP-NaMt is illustrated in Fig. 8. XRD, SEM, and BET analyses confirm that the co-intercalation of CTAB and PVP moderately enlarges the interlayer spacing and partially exfoliates the clay lamellae. This structural modification generates a hierarchical pore network, which significantly enhances the accessibility of adsorption sites and provides the foundation for efficient contaminant uptake (Larraza et al., 2012). Under the optimized condition (pH = 5), XO exists predominantly as the monoanionic species (XO^-) due to partial deprotonation of its sulfonic and phenolic groups. Meanwhile, intercalated CTAB provides uniformly distributed quaternary ammonium sites ($N^+(CH_3)_3$) on both basal surfaces and interlayer

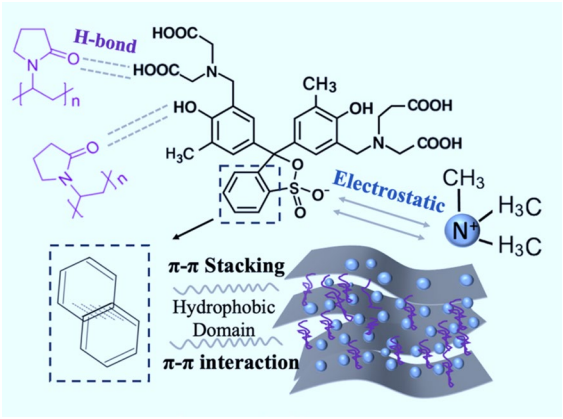

Fig. 8 Proposed mechanism for XO adsorption onto CP-NaMt

regions. The strong electrostatic attraction between these cationic sites and XO^- forms stable ion pairs, which constitute the primary driving force for adsorption. Furthermore, the lactam carbonyl groups (C=O) of PVP and the surface hydroxyl groups (-OH) of the clay act as hydrogen-bond acceptors/donors toward the phenolic/carboxylic groups of XO (Al-Kazragi et al., 2019; Liosi et al., 2017). These directional hydrogen bonds form a network that provides secondary stabilization of the adsorbed complexes. More significantly, the long alkyl chains of CTAB ($(CH_2)_{15}$-CH_3) in combination with PVP backbones create well-defined hydrophobic microdomains within the interlayer spaces, facilitating hydrophobic partitioning of the aromatic XO molecules. Subsequent π-π stacking interactions between adjacent XO molecules in these hydrophobic microdomains further enhance the adsorption stability and retention capacity (Zhu et al., 2019). This multi-mechanistic adsorption system, featuring complementary electrostatic, hydrogen-bonding and hydrophobic/π-π interactions, exhibits both rapid kinetics and high adsorption capacity, making it particularly effective for XO removal from aqueous solutions.

Conclusions

In this study, sequential CTAB-PVP modification successfully converted Na-montmorillonite into a hierarchically porous, partially exfoliated adsorbent (BET 46.5 to 88.7 $m^2 \cdot g^{-1}$). XO uptake was pH-dependent and maximized at pH = 5, with a maximum capacity of 93.85 mg g^{-1} at 30 °C. Within the investigated concentration range, the isotherms were better described by the Langmuir model and adsorption was rapid following a PSO rate law with multi-stage diffusion. Mechanistically, electrostatic pairing of XO^- at CTAB quaternary-ammonium sites dominated was the dominant adsorption mechanism, with additional stabilization via hydrogen bonding and hydrophobic/π–π interactions within organic microdomains. It should be noted that XRD and FTIR analyses were carried out only on pristine Na-Mt and CP-NaMt to confirm structural modification. Thus, the proposed XO adsorption mechanism is mainly inferred from macroscopic behavior, and more detailed spectroscopic characterization of the spent adsorbent is planned for future work. The sorbent was regenerated with ethanol or mild alkaline salt solutions and retained 85–88% of its capacity after five cycles. In summary, CP-NaMt exhibits excellent adsorption toward anionic aromatic dyes in wastewater, highlighting its promise as a low-cost, regenerable adsorbent for practical deployment.

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Author contributions **Yingying Li and Wenxu Wang:** Writing, Conceptualization, Methodology, Validation, Investigation, Resources. **Mengjie Bai:** Methodology, Investigation, Software. **Xia Li and Yongwei Li**: Review and Editing, Supervision, Project administration. **Zhihao Fang and Jing Sang:** Writing, Review and Editing, Supervision, Project administration. **Haiying Liu:** Term, Visualization, Validation, Investigation.

Funding Weifang University of Science and Technology, KJRC2022012

Data availability All the data is included in the manuscript.

Declarations

Conflict of interest The authors declare no conflict of interest.

Ethics approval Not applicable. This study involved no human participants, no animal subjects, and no identifiable personal data. All experiments used commercially available materials and followed standard laboratory safety and waste-disposal regulations.

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